Abed Zahed

RF & Wireless Systems Engineer | Telecom & Wireless Sensing | Engineering Lead

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Nationality: Swedish | Location: Sharjah, UAE

EXPERIENCE

Academic Research Project: Satellite RF Link Budget & Modulation Simulation

American University of Sharjah (Research Collaboration)

06/2026 - Present 📌 *Sharjah, UAE*

- Designed a physics-based link budget model for a GEO S-band satellite link in an ongoing academic research collaboration with the American University of Sharjah; modeled multipath, scintillation, and Faraday rotation referencing ITU-R M.1343 and ECSS-E-ST-50-05C, targeting peer-reviewed publication.
- Implemented CCSDS LDPC adaptive coding and modulation in MATLAB, spanning 3 active QPSK profiles (r=1/2 to r=3/4) and extending evaluation to 8PSK r=7/8 and DVB-S2 16APSK per ETSI EN 302 307-1 for robust link performance across varying channel conditions.
- Ran Monte Carlo simulations generating BER waterfall curves via MATLAB Communications Toolbox; achieved 9.5 dB coding gain vs uncoded QPSK, reducing transmit power by 9x.

Consultant Engineer

Fun Robotics

11/2024 - 08/2025 📌 *Dubai, UAE*

- Managed end-to-end preparation and implementation of technical training courses, coordinating between Fun Robotics and TDRA (UAE Telecom Regulatory Authority) on logistics, scheduling, and program delivery.
- Designed and delivered technical workshops with TDRA, managing curriculum development and coordination with Etisalat Academy.
- Led a team to develop and deliver technical training on wireless communications and RF systems.

Tech Consultant

Mpya Sci & Tech

10/2020 - 08/2024 📌 *Gothenburg, Sweden*

- Worked as engineering consultant across Ericsson and Veoneer client sites via Mpya Sci & Tech's professional-services model.
- Held After-Work Committee Coordinator and Consultant Buddy roles at Mpya, driving team culture initiatives and onboarding new consultants.

SUMMARY

RF and wireless systems engineer with 5+ years across telecom, automotive, and RF research, DSP/RF test development and MRC/MRCx implementation at Ericsson, process engineering and root cause analysis at Veoneer, and antenna sensing research at RISE (3x detection sensitivity gain, published). MATLAB, Python, and FPGA test automation throughout.

LANGUAGES

Arabic	Native
English	Full Professional
Swedish	Elementary (improving)
Spanish	Elementary

LICENSES

✓ EU (Sweden) Driving License: Category B

✓ UAE Driving License

Microwave DSP Config Developer (Consultant)

Ericsson

03/2022 - 12/2023 📌 *Gothenburg, Sweden*

- Implemented MRC and novel MRCx combining on XPIC channel modes for Ericsson MiniLink; achieved 2.7 dB signal power gain.
- Extended modulation profile work beyond MRC/MRCx combining to cover higher-order QAM schemes up to 32-QAM.
- Developed MATLAB and Python test bench for MiniLink validating SISO, XPIC, and MIMO channel configurations end-to-end.

Process Engineer (Consultant)

Veoneer / Magna International

11/2020 - 01/2022 📌 *Vårgårda, Sweden*

- Improved machine performance from 65% to 98% by leading a team of 3 optics engineers in troubleshooting, root cause analysis, and documentation, part of the camera production line's industrialization and ramp-up to full-scale manufacturing.
- Directed incident and problem resolution for camera production; coordinated cross-functional investigations with production, engineering, and maintenance teams.
- Built Power BI dashboards and Python pipelines to track production efficiency and process capability KPIs; adopted org-wide for yield monitoring.

Space Systems Research (M.Sc. Program)

Onsala Space Observatory / Chalmers University

2018 - 2019 📌 *Onsala, Sweden*

- Completed research training at Onsala Space Observatory covering satellite communications, link budgets, and space systems engineering as part of M.Sc. in Wireless, Photonics and Space Engineering at Chalmers.
- Studied orbital mechanics and satellite platform design, radar systems with Doppler processing and FMCW waveforms, and satellite positioning/GNSS with pseudorange-based differential positioning.
- Operated GNSS tracking, satellite positioning, and antenna dish control systems in an active deep space observatory, generating the raw datasets used for deep-space data processing and analysis.

SKILLS

CORE COMPETENCIES

- Technical Training & Knowledge Transfer
- Regulatory Compliance (ETSI / ANSI / IEC)
- Cross-functional Team Leadership
- TDRA / UAE Telecom Regulatory Coordination

TELECOM & ENGINEERING

- Machine Learning (ANN, SVM , RF sensing classification)
- SQL (data queries for analysis)
- Digital Modulation and Coding (QPSK, 8PSK, DVB-S2, CCSDS LDPC)
- Wireless & Microwave Systems (LTE/5G, Radio Links)
- RF Testing, Validation & Performance Analysis
- RF Lab Equipment and Instrumentation (VNA, Spectrum Analyzer, Signal Generators, Network Analyzers)
- Circuit and PCB Design (RF antenna PCBs, fractal antennas, diagnostic test setups; Altium)
- RF Front-End Systems (LNAs, PAs, mixers, filters, T/R modules; systems-level design)
- Digital Signal Processing (DSP)
- MIMO / XPIC / SISO channel modeling

TOOLS & PLATFORMS

- Power BI
- Python (test automation, data analysis)
- MATLAB
- Electronics Testing and Diagnostics (RF systems, channel emulation, performance validation)
- FPGA-based channel emulation
- Simulink
- HFSS
- ADS, CST Microwave Studio (EM simulation)

Master's Thesis: RF Researcher

RISE Research Institutes of Sweden

02/2019 - 06/2019 *Borås / Gothenburg, Sweden*

- Increased detection sensitivity by 3x via patch antenna research for transformer partial discharge monitoring.
- Designed and tested a prototype patch antenna using VNA, spectrum analyzer, and signal generators; calibrated test setup demonstrated performance comparable to standardized measurement setups.
- Measured and characterized radiation patterns for the patch antenna prototype; validated gain, return loss, and 50Ω impedance matching across the UHF frequency range.

Robotics Engineer

Fun Robotics

06/2016 - 08/2017 *Dubai, UAE*

- Taught wireless communication fundamentals, protocol configuration, embedded systems, and RF signal concepts to students age 7-20.
- Designed comprehensive teaching curricula and supervised student project work across robotics and software platforms for school- and university-level students; adopted as educational resources for the academic industry.
- Designed and manufactured robotic solutions as part of the product development team.

Visiting Researcher

American University of Sharjah

01/2015 - 10/2015 *Sharjah, UAE*

- Co-authored 4 IEEE journal and conference publications in RF sensing and diagnostic antenna design.
- Designed PCB-based RF antennas with 50Ω coaxial feeds; applied transmission line theory for impedance matching and optimized RF circuit feed networks.
- Simulated 4th-order Hilbert fractal antenna designs in HFSS; characterized 2D and 3D radiation patterns to verify semi-spherical coverage and directivity at VHF/UHF (0.3-6 GHz).

EDUCATION

M.Sc. in Wireless, Photonics, and Space Engineering

Chalmers University of Technology

2017-2019 *Gothenburg, Sweden*

B.Sc. in Electrical Engineering (Minor: Computer Engineering)

American University of Sharjah

2011-2015 *Sharjah, UAE*

Includes CCNA (Cisco Certified Network Associate) introductory coursework, familiar with Cisco Packet Tracer simulation tools

KEY ACHIEVEMENTS

♦ MRC/MRCx Implementation

Led implementation on Ericsson MiniLink products achieving 2.7 dB gain in received signal power, effectively doubling received power.

♦ UAE Telecom Training Delivery

Designed and delivered technical training with UAE TDRA in partnership with Etisalat Academy, supporting regulatory capability building.

♦ Production Performance Turnaround

Improved camera production line performance from 65% to 98% at Veoneer/Magna by leading a team of 3 optics engineers through root cause analysis and corrective action.

♦ RF Sensing Research

Increased detection sensitivity 3x via patch antenna research at RISE Sweden; co-authored 4 IEEE publications in RF sensing and diagnostic antenna design.

♦ Operational Analytics

Developed Power BI dashboards adopted org-wide at Veoneer for production yield monitoring and Python-automated data analysis pipelines.